

EXPERIMENT 15: FEED FORWARD NEURAL NETWORK

AIM

To implement a simple **Feed Forward Neural Network (FFNN)** in Python for performing classification.

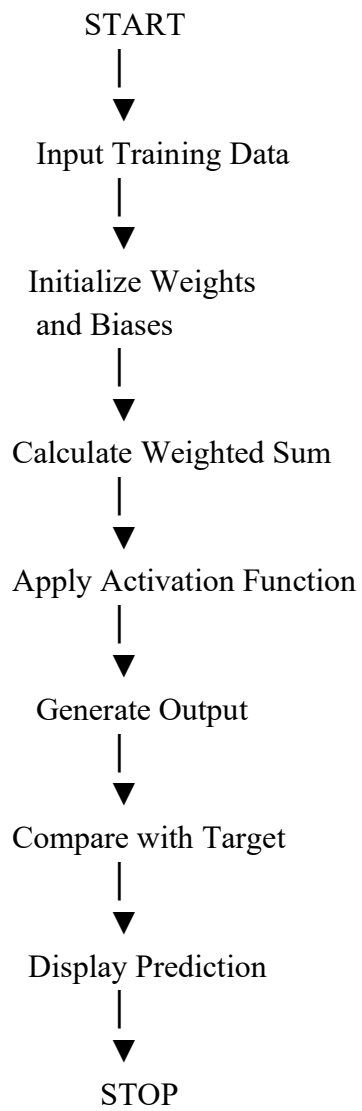
OBJECTIVE

- To understand the basic structure of a neural network.
- To understand input, hidden, and output layers.
- To implement forward propagation.
- To classify input data using a neural network.

ALGORITHM

1. Start the program.
2. Initialize the input data and target values.
3. Initialize weights and biases.
4. Calculate the weighted sum of inputs.
5. Apply the activation function.
6. Pass the output to the next layer.
7. Calculate the final output.
8. Compare the predicted output with the target.
9. Display the prediction.
10. Stop.

FLOWCHART



PYTHON PROGRAM

```
import math
def sigmoid(x):
    return 1 / (1 + math.exp(-x))
# Input data
X = [1, 0]
# Weights and bias
w1 = 0.5
w2 = 0.5
b = -0.2
# Feed forward
z = X[0] * w1 + X[1] * w2 + b
output = sigmoid(z)
print("Input:", X)
print("Output:", round(output, 3))
if output >= 0.5:
    print("Predicted Class: 1")
else:
    print("Predicted Class: 0")
```

SAMPLE OUTPUT



RESULT

Thus, the **Feed Forward Neural Network** was successfully implemented in Python and the input data was classified.

CONCLUSION

The experiment demonstrates the basic working of a Feed Forward Neural Network using **weights, bias, weighted sum, and the sigmoid activation function.**